

Final Project Overview: Secure Access

Overview

In this course, you learned the crucial skills to tackle modern, evolving digital threats and challenges. From threat detection to prevention strategies, you learned to safeguard your organizations and systems against data breaches. In this project, you will explore a real-world-inspired scenario, identify potential security vulnerabilities, and provide recommendations to strengthen the organization's overall security posture.

This final project comprises three tasks:

1. Evaluate the organization's existing security infrastructure
2. Create a multifactor authentication (MFA) plan
3. Evaluate the organization's existing physical security measures

General instructions

The final project consists of three tasks, each containing two questions, contributing to a total of 15 points. Carefully read through the scenarios and the accompanying tasks. Then, critically analyze each question and provide well-considered responses to demonstrate your understanding and proficiency.

To complete this project, you must answer all the questions and submit the assignment as proof of completing the final project's requirements. You can pause and resume the submission process based on your schedule, giving you the flexibility to complete this project at your own pace. Your submission will be auto-graded using an AI tool.

Scenario for Task 1 and Task 2

You are an IT Security Analyst at TechSolutions Inc. The company has recently experienced a data breach due to compromised credentials.

TechSolutions Inc. currently employs the following security infrastructure:

- The traditional access control methods include:
 - Username and password sign-in for each system and application.
 - SSH keys to access remote servers.
 - Smart cards for secure physical and digital access across multiple company locations and the corporate VPN.
- The resources that employees require access to:
 - **Corporate email system:** Essential for day-to-day communication, both internally and with external stakeholders.
 - **Internal databases:** Contains sensitive information such as client details, project data, and financial records.
 - **Cloud storage services:** Used for storing and sharing documents and collaborative work.
 - **HR systems:** Incorporates personal information of employees, benefits administration, and recruitment data.
 - **Customer relationship management (CRM) software:** Critical for sales, marketing, and customer service operations.
 - **Project management tools:** Used for tracking progress, assigning tasks, and optimizing workflow.
 - **Development environments:** Required for software development, testing applications, and managing code repositories.
 - **Company intranet and knowledge bases:** Provides access to company policies, training materials, and internal communication platforms.

Task 1 Instructions

This task comprises two questions and carries 6 points. Thoroughly review the scenario above and answer the questions.

Review the security infrastructure currently deployed at TechSolutions Inc. This task will require you to leverage your understanding of cybersecurity principles and apply critical thinking to assess and enhance the security infrastructure of TechSolutions Inc.

Task 2 instructions

This task includes two questions with a total value of 3 points.

Consider the various authentication methods you learned in this course. Your task is to revamp the access control system at TechSolutions Inc. by incorporating multifactor authentication (MFA) to safeguard against future unauthorized access.

Scenario for Task 3

TechSolutions Inc. currently employs the following physical security measures:

- **Reception and lobby:** The entrance features a reception desk that is always manned. The receptionist takes the names of each visitor and issues temporary visitor badges.
- **Employee workspaces:** The open-plan workspaces for employees are accessible from two directions. One entrance, located near the receptionist's desk, leads to the front of the workspace area, while another door at the back opens into the employee parking lot. Both entrances are clearly marked with signs reading "Authorized Personnel Only." The receptionist locks the front door, and the last person leaving for the day locks the back door.
- **Meeting rooms and conference halls:** Larger meeting spaces are adjacent to the shared workspace. Each of these spaces has doors that can be locked. Reservations for these spaces can be made through a central calendar system.
- **Data centers and server rooms:** These critical areas are equipped with lockable doors, with access restricted to the IT and janitorial staff. Additionally, these rooms are fitted with a thermostat and humidity sensor for monitoring purposes.
- **File storage:** File cabinets with locking mechanisms are strategically located throughout the office space to ensure sensitive documents are secure and accessible only by authorized personnel.

- **Parking lot:** The employee parking lot is enclosed by a fence with only one entry point. Parking stickers are required to access this parking lot.
- **Common areas and facilities:** Breakrooms and restrooms are adjacent to the common work area. The doors leading to these areas do not have locks.

Task 3 Instructions

This task consists of two questions totaling 6 points. Thoroughly review the scenario and answer the questions.

Review the physical security measures currently deployed at TechSolutions Inc. This task will require you to leverage your understanding of cybersecurity principles and apply critical thinking to assess and enhance the physical security measures at TechSolutions Inc.



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